

Literature answer to a uniform-boundedness problem in limit operators

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Status. This is a **literature already answered** packet for Open Problem 8 of Chandler-Wilde–Lindner, not a new proof packet.

Source problem. Chandler-Wilde–Lindner, *Limit Operators, Collective Compactness, and the Spectral Theory of Infinite Matrices* (arXiv:1005.0166), end with a list of open problems. Open Problem 8 asks whether the uniform boundedness condition in Theorem `prop_rich_invinf`(iii) is redundant for $p \in (1, \infty)$. In parsed source lines 7306–7312, the paper says that the redundancy was known for $p \in \{0, 1, \infty\}$, while for the remaining cases $p \in (1, \infty)$ the question remained open.

Supporting answer. Lindner–Seidel, *An Affirmative Answer to a Core Issue on Limit Operators* (arXiv:1401.1300), directly addresses the same issue. Its abstract says that for rich band-dominated operators the known Fredholm criterion required all limit operators to be invertible with uniformly bounded inverses, and that the paper shows the uniform-boundedness condition is redundant.

The introduction identifies the question as the “big question”: whether the operator spectrum of a rich operator is automatically uniformly invertible if it is elementwise invertible. Lines 265–267 of the parsed source state that the aim is to solve this problem in general by showing that the condition is redundant. The precise statements are Corollary 3.7 and Theorem 3.8. Corollary 3.7 proves that if A is rich and band-dominated and all $B \in \sigma_{\text{op}}(A)$ are invertible, then the norms $\|B^{-1}\|$ are uniformly bounded. Theorem 3.8 then gives the simplified criterion:

$$A \in \mathcal{A}_p^{\$}, \quad A \text{ is } \mathcal{P}\text{-Fredholm} \iff \text{all limit operators of } A \text{ are invertible,}$$

for A acting on $\ell^p(\mathbb{Z}^N, X)$ with $p \in \{0\} \cup [1, \infty]$.

Identification and scope. This answers the source’s Open Problem 8 affirmatively, and in particular covers the previously open range $1 < p < \infty$. The answer is literature, not new work by this run. The supporting theorem keeps the rich and band-dominated hypotheses; Section 4 of the supporting paper gives examples showing these hypotheses cannot simply be dropped.

Search evidence. The cheap run indexes were searched for arXiv:1005.0166, arXiv:1401.1300, limit operators, collective compactness, Fredholmness, essential spectrum, and the uniform boundedness condition. No existing run packet recorded this exact answer. External arXiv search for phrases around “uniform boundedness condition”, “limit operators”, and “redundant” located arXiv:1401.1300 as the decisive later source.

References

- [1] S. N. Chandler-Wilde and M. Lindner, *Limit Operators, Collective Compactness, and the Spectral Theory of Infinite Matrices*, arXiv:1005.0166.
- [2] M. Lindner and M. Seidel, *An Affirmative Answer to a Core Issue on Limit Operators*, J. Funct. Anal. 267(3) (2014), 901–917; arXiv:1401.1300.